

Multiple Choice (10 marks)

1. Which is the largest asymptotically?

- (a) $O(1)$
- (b) $O(n)$
- (c) $O(n^2)$
- (d) $O(\log n)$

2. In Python 3, what is the output of `['Hi!'] * 4`?

`'Hi!', 'Hi!', 'Hi!', 'Hi!'`

`'Hi!' * 4`

- (a) Error
- (b) None of the above.

3. Which is a valid expression in Python 3.5?

- (a) `sort('ab')`
- (b) `sorted('ab')`
- (c) `"ab".sort()`
- (d) `1/0`

4. Which types of functions grow the slowest?

- (a) $O(\log e^N)$
- (b) $O(\log N)$
- (c) $O(\log \log N)$
- (d) $O(N)$

5. Let `l = [1,2,3,4]`. What is `max(l)` in Python 3?

- (a) 1
- (b) 2
- (c) 3
- (d) 4

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the function `cmp(list)` determines the count of items in the list.
7. True or False: An Internet Protocol (IP) address is assigned to a device when it is connected to the Internet.
8. True or False: The boolean expression `a[i] == max || !(max != a[i])` simplifies to `a[i] != max`.
9. True or False: In Python 3, the `strip([chars])` function removes all leading and trailing whitespace from a string.
10. True or False: In Python 3, using negative indexing on a string will always result in an error.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find t-nth term of arithmetic progression.
12. Write a function to extract all the adjacent coordinates of the given coordinate tuple.

End of Paper